NLU course project - Part 2A and Part 2B

Emanuele Ricca (258437)

University of Trento

emanuele.ricca@studenti.unitn.it

1. Part 2.A – Joint Intent and Slot Model from Scratch

1.1. Introduction

In this exercise, I adapted the Transformer architecture to a multi-task setting for the ATIS dataset. The model predicts both the intent of the utterance and the slot label for each token. I followed the same incremental strategy as in the language-model task, but now the evaluation criteria are intent accuracy and slot F1 instead of perplexity.

1.2. Implementation details

I built a custom preprocessing pipeline with a vocabulary, dataset wrapper, and collate function that pads sequences and preserves lengths for batched training. The model produces two outputs: one sequence-level intent classifier and one slot classifier over the token representations. I first tuned the baseline hyperparameters, then adjusted model width, number of heads, depth, and feed-forward size one by one. I also added dropout before the final classification layers to regularize the joint model.

1.3. Results

The final report will summarize the best configuration for the joint model and describe how the accuracy/F1 balance changed as capacity and regularization were modified.

2. Part 2.B – Fine-Tuning GPT2 and BERT

2.1. Introduction

This exercise fine-tunes pre-trained Transformer models for the same ATIS multi-task problem. I adapted both GPT2 and BERT to perform intent classification and slot filling, while accounting for their different architectures and tokenization strategies. The main challenge was aligning the slot supervision with subtokenized inputs.

2.2. Implementation details

For GPT2, I reused the decoder-only setup and adapted the final representations for both intent and slot prediction. For BERT, I used the encoder representations directly and built a classification head on top. In both cases, I handled sub-tokenization by aligning labels to the tokenized sequence so that slot tags remain consistent after wordpiece splitting. Training optimized the two task losses jointly, and the evaluation protocol used intent accuracy together with CoNLL-style slot F1.

2.3. Results

The final report will include a comparison between the fine-tuned GPT2 and BERT models, with comments on which architecture handled the joint task more effectively and under which tokenization constraints.